روش‌های پیشرفته نمونه برداري در جنگل

الگوهای نمونه‌برداری

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مدل مفهومي نمونه برداري

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Simple Random Sampling



$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$s_x^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

$$E = \pm t_{\frac{\alpha}{2}, df} s_{\bar{x}}$$

$$CI = \bar{x} \pm E$$

$$E\% = 100 \times \frac{E}{\bar{x}}$$

Simple Random Sampling



$$N = \frac{size\ of\ population}{size\ of\ sampling\ unit}$$

$$Total = N \times \bar{x}$$

$$S_{\overline{Total}} = N \times S_{\bar{x}}$$

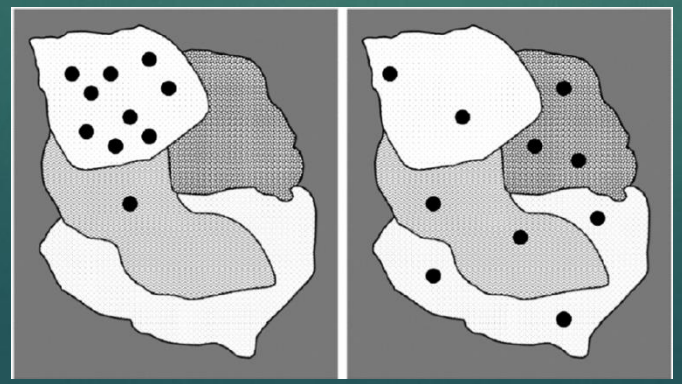
$$CI_{Total} = Total \mp tS_{\overline{Total}}$$

Advantages and disadvantages of SRS

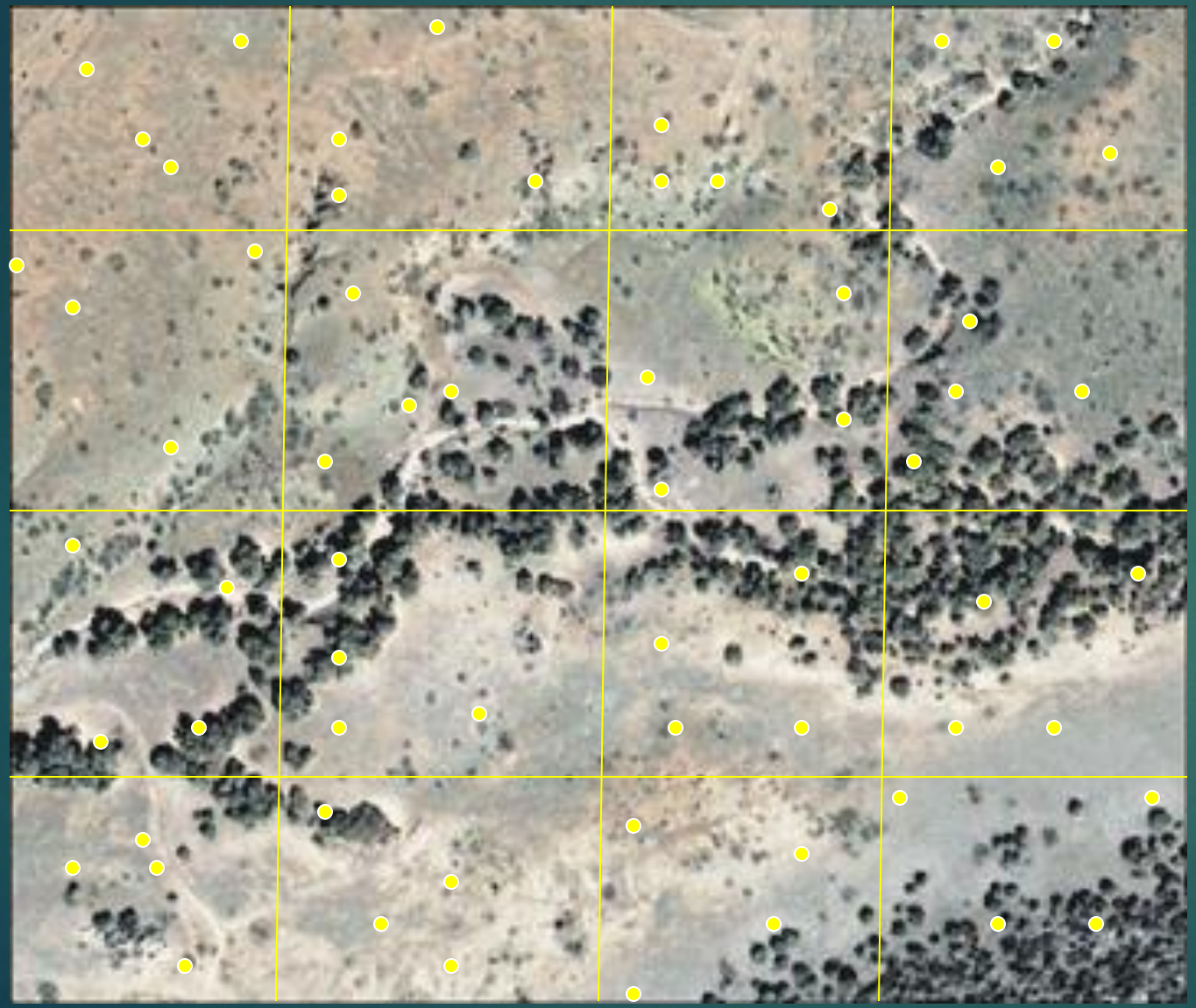
- Strict adherence to the principles of simple random selection guarantees unbiased and consistent estimates of population parameters and their standard errors.

- A SRS design often requires a surprisingly large investment in organization, checking, and location of the samples,

- Also, through random selection an irregular spatial distribution of sample locations may result, so the population as a whole is not uniformly represented



Block Sampling



$$\bar{x}_j = \frac{\sum_{i=1}^{k_j} x_{ij}}{k_j}$$

$$\bar{\bar{x}} = \frac{\sum_{j=1}^{M} \bar{x}_j}{M}$$

$$S_{x_j} = \sqrt{\frac{\sum_{i=1}^{k_j} x_{ij}^2 - \frac{(\sum_{i=1}^{k_j} x_{ij})^2}{n}}{k_j - 1}}$$

$$S_{\bar{x}} = \sqrt{\frac{\sum_{j=1}^{M} s_{x_j}^2}{M \times n}} \left(\frac{N-n}{N}\right)$$

$$E = \pm t_{\frac{\alpha}{2}, df} \, s_{\bar{x}}$$

Systematic Random Sampling



$$S_{\bar{x}} = \sqrt{\frac{\sum_{j=1}^{M}\sum_{i=1}^{n_j}(x_{ij} - x_{(i+1)j})^2}{2n\sum_{j=1}^{M}(n_j - 1)}}$$

$$E = \pm t_{\frac{\alpha}{2}, df} s_{\bar{x}}$$

$$CI = \bar{x} \pm E$$

$$E\,\% = 100 \times \frac{E}{\bar{x}}$$

$$S_{\overline{Total}} = N \times S_{\bar{x}}$$

$$CI_{Total} = Total \mp tS_{\overline{Total}}$$

Advantages and disadvantages of Systematic Sampling

The efficiency is generally better than using SRS

It is possible to derive unbiased estimators of totals and means from this sample but not of variance (Thompson 1992)

It is generally impossible to provide unbiased estimators of the variances when systematic sampling is used. Attempts have been undertaken to find estimators with little bias and low variance (Bellhouse 1985;Wolter 1985; Sherman 1996). The most commonly used approach is based on the assumption that a systematic sample is equivalent to a random sample

SRS estimators of standard errors applied to estimates from systematic sampling are usually conservative; they overestimate, on average, the actual error. An overestimation of about one third is not unusual, but more extreme results have been reported (Hartley 1966; Bellhouse 1988; Stehman 1992)

Approaches

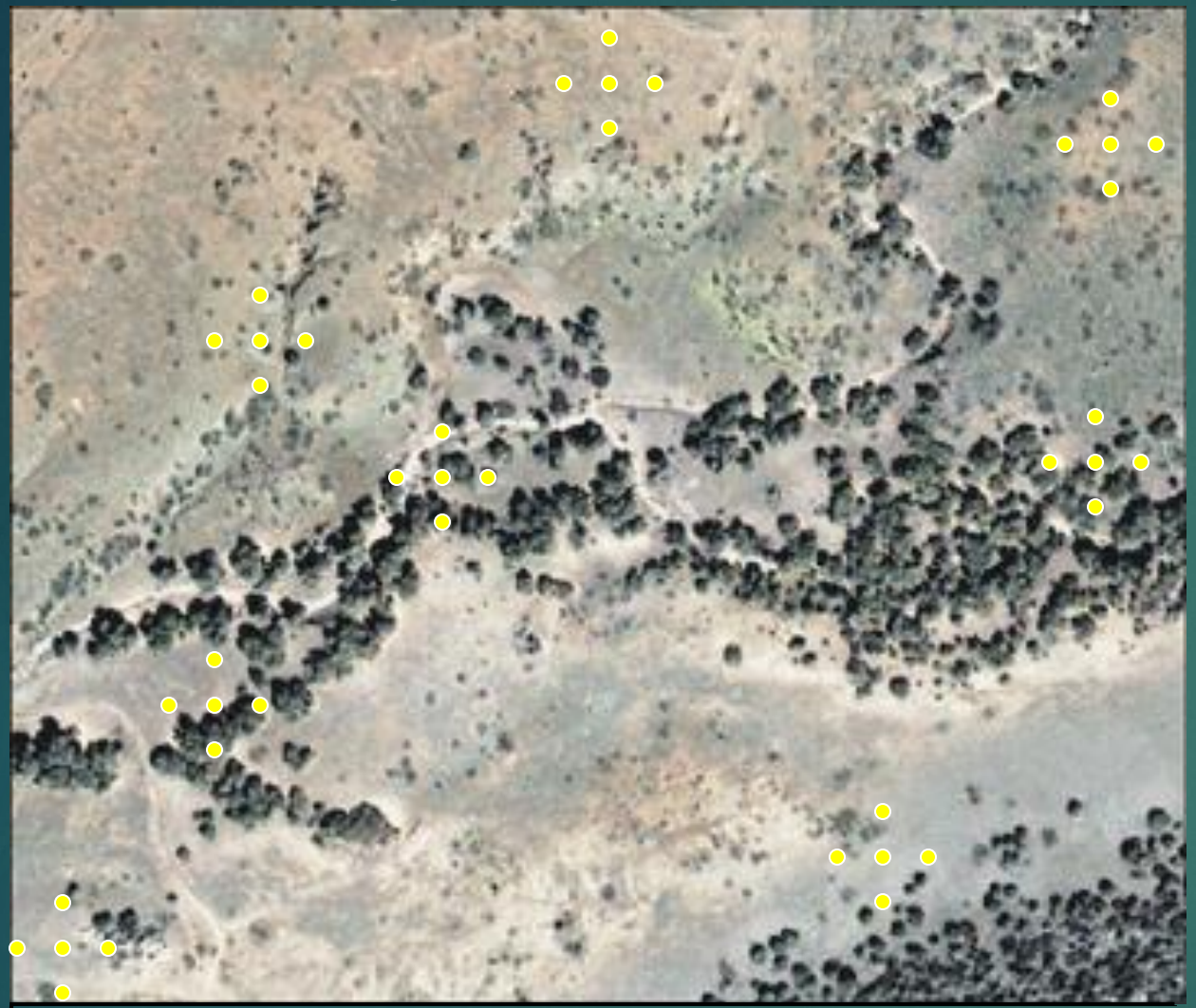
Jessen (1942), Yates (1949, 1981), and Cochran (1977) suggested a procedure that involves grouping of pairs of adjacent sample units (clusters). Each grouped pair of units is considered as a single stratum in a stratified sampling scheme, which allows the computation of a within-stratum variance, an estimate needed for the purpose of correcting the "inflated" SRS variance estimate. This procedure leads to corrected variance estimates that are biased, the bias being either positive or negative.

Yates (1981) suggested for the situation of two dimensional sampling to combine sample units into blocks of 2×2 units. Each block is then considered as a stratum in stratified random sampling.

Instead of using squared residuals for computing a sampling variance, von Neumann et al. (1941) proposed using the sum of squared differences between adjacent sample units. This method is known as the method of squared differences (Ekström and Sjöstedt de Luna 2004; Stevens and Olsen 2004).

The ideas of Neumann and Yates have been developed further in the form of NN estimators of local expectations and estimating the variance from the squared differences between the sample values and NN "predictions" (Ekström and Sjöstedt de Luna 2004; Stevens and Olsen 2004).

Cluster Sampling



$$\bar{x}_j = \frac{\sum_{i=1}^{k_j} x_{ij}}{k_j}$$

$$\bar{x} = \frac{\sum_{j=1}^{M} \bar{x}_j}{M}$$

$$S_x = \sqrt{\frac{\sum_{j=1}^{m}(\sum_{i=1}^{k_j} x_{ij})^2 - 2\times\bar{x}\sum_{j=1}^{m}[(\sum_{i=1}^{k_j} x_{ij})\times k_j] + \bar{x}^2 \times \sum_{j=1}^{m} k_j}{k(m-1)}}$$

$$S_{\bar{x}} = \sqrt{\frac{\sum_{j=1}^{m} s_{x_j}^2}{k \times m}}$$

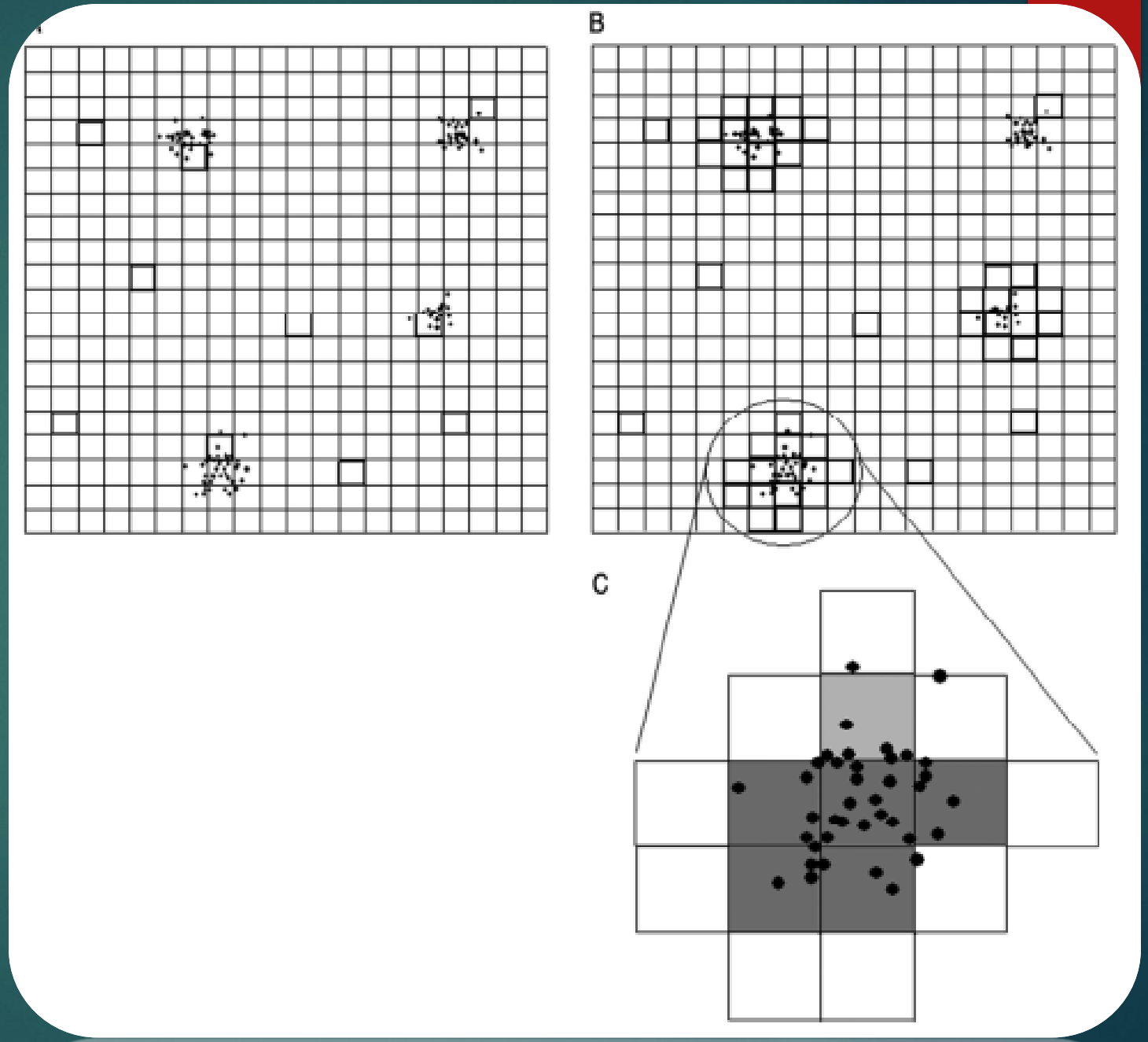
Some points

From the viewpoint of expense, such grouping of sampled elements/units is justifiable. However, adding two or more “extra” units/elements at each sample location does not necessarily mean that the sample size can be reduced by a factor equal to the number of added elements/units at each sample location.

The trade-off between the number of elements/units sampled per sample location and the number of sample locations to visit in order to achieve a target precision on estimates of population parameters depends on the distribution of the variance of attribute values across spatial and temporal scales.

In general, the less the variance of the attribute value is within a cluster, the less is the efficiency of clustering sample observations relative to a SRS of individual units/elements.

Adaptive Cluster Sampling



For an initial sample of primary units, the modified ACS HH estimator is:

$$\widehat{\mu}_{HH} = \frac{1}{Mn_1} \sum_{k=1}^{K} \frac{b_k y_k^*}{x_k} \tag{16}$$

where b_k is the number of times network k is intersected by the initial sample. Thompson (1991a) showed $\widehat{\mu}_{HH}$ is unbiased with variance:

$$Var[\widehat{\mu}_{HH}] = \frac{\sigma_W^2}{n_1}\left(1 - \frac{n_1}{N}\right) \tag{17}$$

where $\sigma_W^2 = \frac{1}{N-1}\sum_{i=1}^{N}(w_i - \mu)^2$, $w_i = \frac{1}{M}\sum_{k=1}^{K}\frac{I_{ik}y_k^*}{x_k}$, and I_{ik}=1 if the ith primary unit intersects the kth network, and I_{ik}=0 otherwise. An unbiased estimator of $Var[\widehat{\mu}_{HH}]$ is:

$$\widehat{Var}[\widehat{\mu}_{HH}] = \frac{s_W^2}{n_1}\left(1 - \frac{n_1}{N}\right) \tag{18}$$

where $s_W^2 = \frac{1}{n_1-1}\sum_{i=1}^{n_1}(w_i - \widehat{\mu}_{HH})^2$.

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